Fundamentals of Electric Circuits

电路基础

Lecture11: Frequency Response

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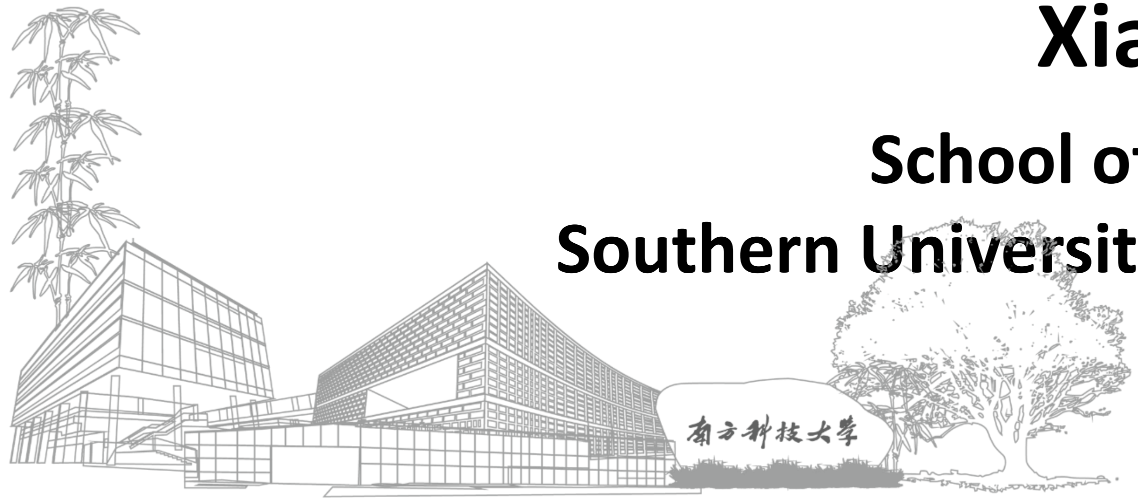


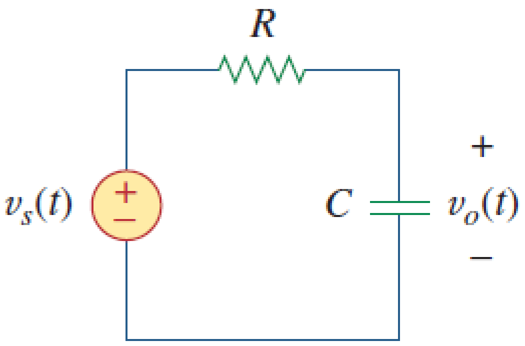
Outline

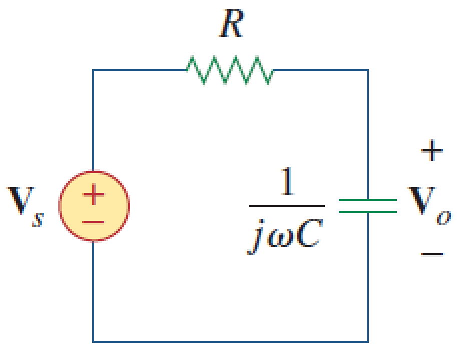
- Introduction
- Transfer Function
- Series Resonance
- Parallel Resonance
- Passive Filters
- Active Filters

Introduction

- We have learned how to calculate the steady-state response of a circuit to AC signals/sources using the technique of phasor representation → The signal frequency is assumed to be constant and the impedance of elements are fixed
- In this lecture, we are going to learn the behavior of a circuit to varying signal frequency
- If we let the amplitude of the sinusoidal source remain constant and vary the frequency, we obtain the circuit's frequency response (频率响应)

> The frequency response of a circuit is the variation in its behavior with change in signal frequency.

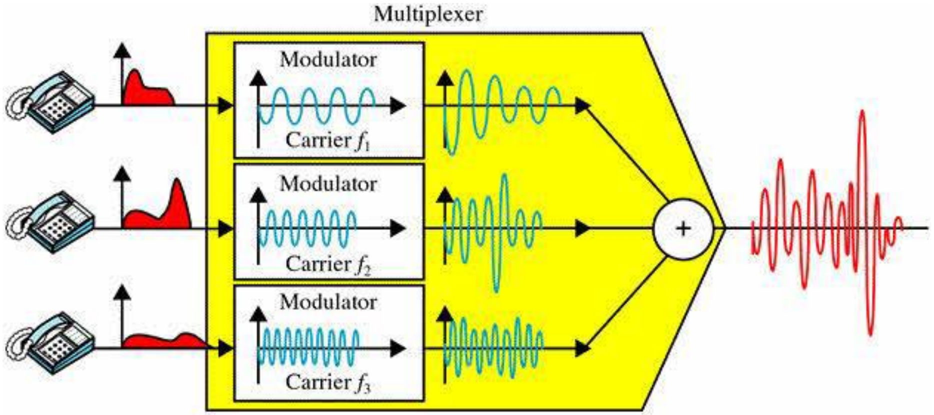




Frequency Response are Crucial

- HiFi (High-Fidelity, 高保真) Audio System, uniform gain and time delay

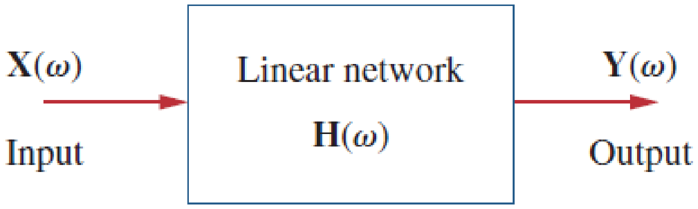
- Frequency division multiplexing



- FM radio receiver

Transfer Function

- One useful way to analyze the frequency response of a circuit is the concept of the transfer function (传输函数)
- The transfer function $\mathbf{H}(\omega)$ of a circuit refers to the frequency dependent ratio of a phasor output $\mathbf{Y}(\omega)$ (an element voltage or current) to a phasor input $X(\omega)$ (source voltage or current)



Magnitude response
幅频响应

$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} = |H(\omega)|\, e^{j\varphi(x)}$$

phase response
相频响应

$$\mathbf{H}(\omega) = \text{Voltage gain} = \frac{\mathbf{V}_o(\omega)}{\mathbf{V}_i(\omega)}$$

$$\mathbf{H}(\omega) = \text{Current gain} = \frac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)}$$

Voltage/current gain
Unitless (无量纲)

转移阻抗

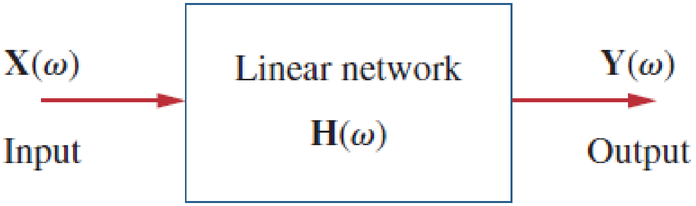
$$\mathbf{H}(\omega) = \text{Transfer Impedance} = \frac{\mathbf{V}_o(\omega)}{\mathbf{I}_i(\omega)}$$

转移导纳

$$\mathbf{H}(\omega) = \text{Transfer Admittance} = \frac{\mathbf{I}_o(\omega)}{\mathbf{V}_i(\omega)}$$

Transfer Function: Zeros and Poles

- The transfer function $\mathbf{H}(\omega)$ can be expressed as the ratio of numerator polynomial **N**(ω) and denominator polynomial **D**(ω)



$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} \qquad \mathbf{H}(\omega) = \frac{\mathbf{N}(\omega)}{\mathbf{D}(\omega)}$$

- Zeros (零点) are where the transfer function goes to zero, and poles (极点) are where it goes to infinity
- The *zeros* usually represents as $j\omega = z_1, z_2, \ldots$. Similarly, the *poles* are usually represented as $j\omega = p_1, p_2, \ldots$
- To avoid complex algebra, it is expedient to replace $j\omega$ temporarily with s when working with $\mathbf{H}(\omega)$ and replace s with $j\omega$ at the end

Example: Transfer Function

- Q: Obtain the transfer function V_o/V_s and its frequency response. Let $v_S = V_m \cos\omega t$

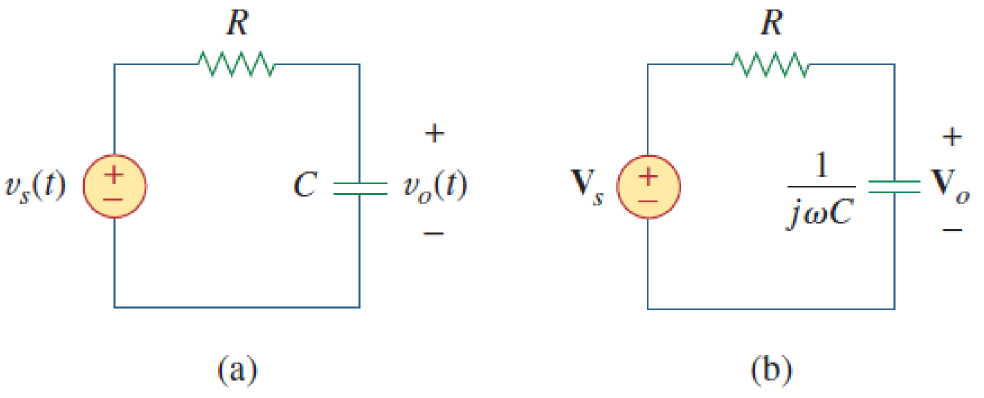
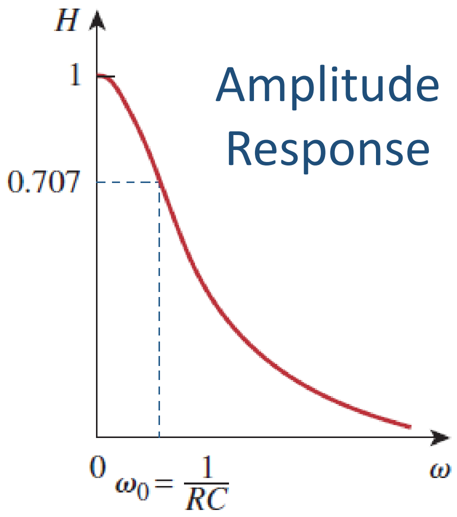
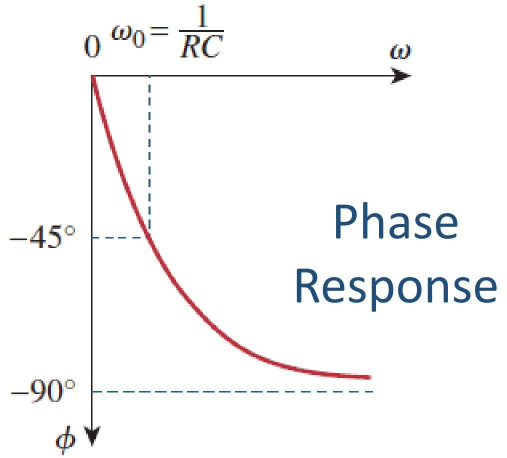


Figure 14.2
For Example 14.1: (a) time-domain RC circuit, (b) frequency-domain RC circuit.





Solution:
The frequency-domain equivalent of the circuit is in Fig. 14.2(b). By voltage division, the transfer function is given by

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$$

$$H = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}, \qquad \phi = -\tan^{-1}\frac{\omega}{\omega_0}$$

TABLE 14.1
For Example 14.1.

ω/ω_0	H	ϕ	ω/ω_0	H	ϕ
0	1	0	10	0.1	−84°
1	0.71	−45°	20	0.05	−87°
2	0.45	−63°	100	0.01	−89°
3	0.32	−72°	∞	0	−90°

The Decibel Scale

- It is not always easy to get a quick plot the magnitude and phase of the transfer function as we did aforementioned → To use the bode plots, but we should take care of two important issues: 1) the use of logarithms and 2) decibels in expression gain
- The decibel (symbol: dB, 分贝) is a unit of measurement used to express the ratio of one value of a physical property to another on a logarithmic scale. It can be used to express a relative change or an absolute value related to a reference
- Historically, the decibel is used to measure the ratio of two levels of power

$$G_{\mathrm{dB}} = 10 \log_{10} \frac{P_2}{P_1}$$

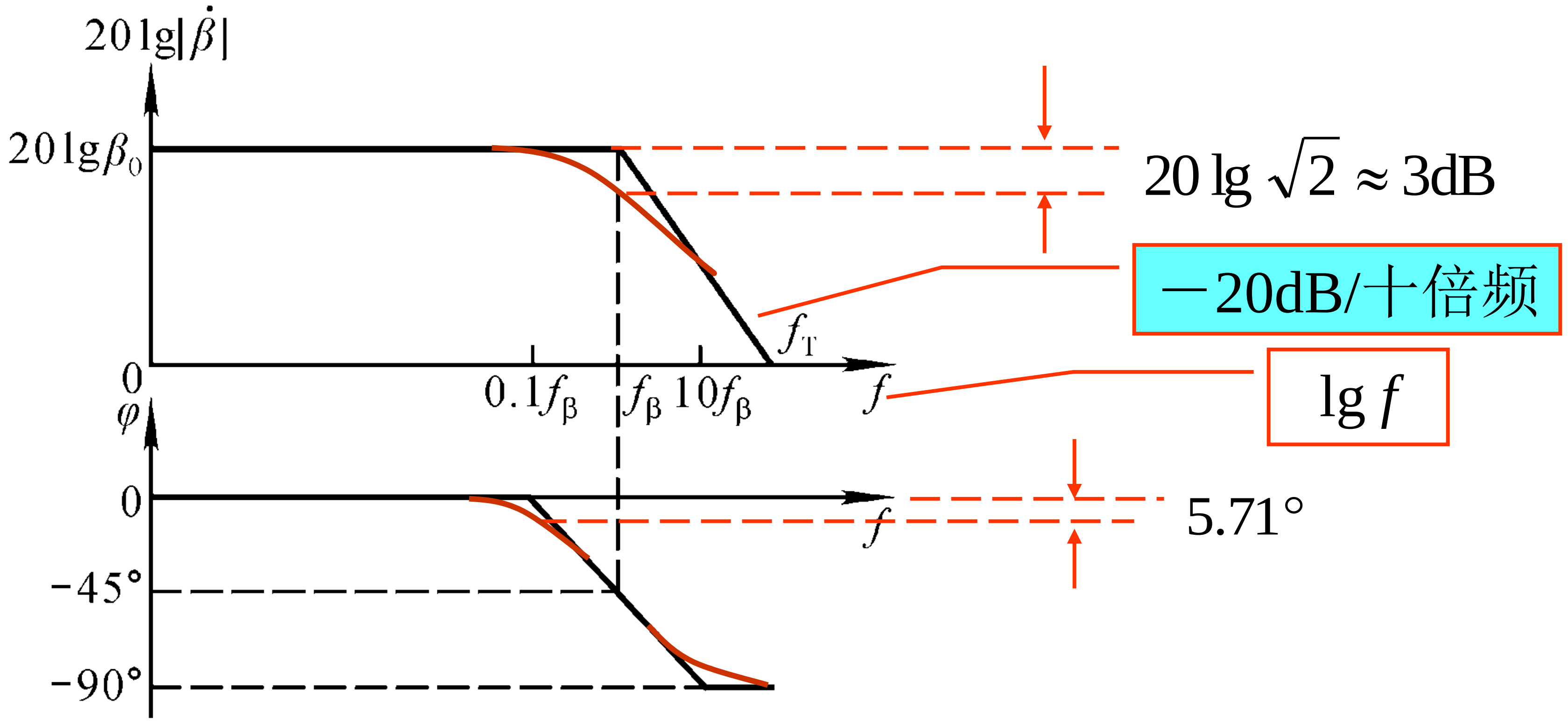
  - When $P_1 = P_2$, there is no change in power and the gain is 0 dB
  - If $P_2 = 2P_1$, the gain is ~3 dB
  - If $P_2 = 0.5P_1$, the gain is ~-3 dB

- Alternatively, the gain can be expressed in terms of voltage or current ratio

$$G_{\mathrm{dB}} = 20 \log_{10} \frac{V_2}{V_1} \qquad G_{\mathrm{dB}} = 20 \log_{10} \frac{I_2}{I_1}$$

Bode Plots

- Bode plots (波特图) are semilog plots (半对数坐标) of the magnitude (in decibels) and phase (in degrees) of a transfer function versus frequency
- One problem with the transfer function is that it needs to cover a large range in frequency → Plotting the frequency response on a semilog plot (where the x axis is plotted in log form) makes the task easier



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- **Series Resonance**
- **Parallel Resonance**
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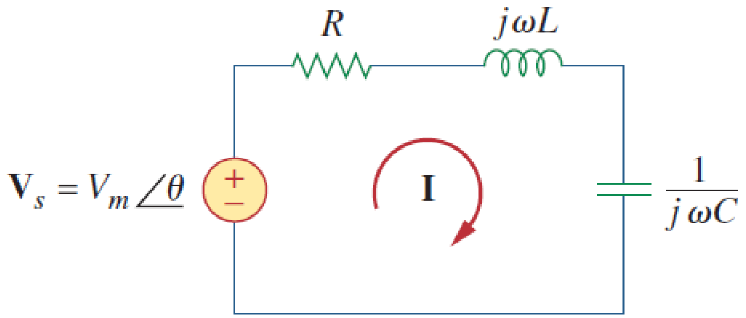
Series Resonance (I)

- The most prominent feature of the frequency response of a circuit may be the sharp peak (or resonant peak, 谐振峰) exhibited in its amplitude characteristic
- Resonance (谐振) is a condition of a circuit involving an *RLC* circuit in which the capacitive and inductive reactance are equal in magnitude, resulting in zero imaginary part in its impedance (or a purely resistive impedance)
- The input impedance is



$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C} \quad \text{or} \quad \mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

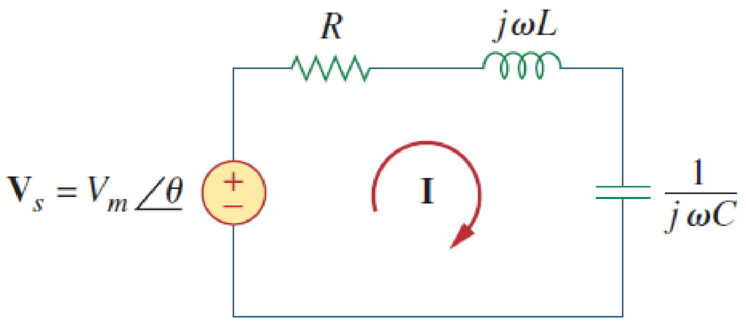
- Resonance occurs when $\mathrm{Im}(\mathbf{Z}) = \omega L - \dfrac{1}{\omega C} = 0$

- The corresponding frequency is called resonant frequency

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}\ \mathrm{rad/s} \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}\ \mathrm{Hz}}$$

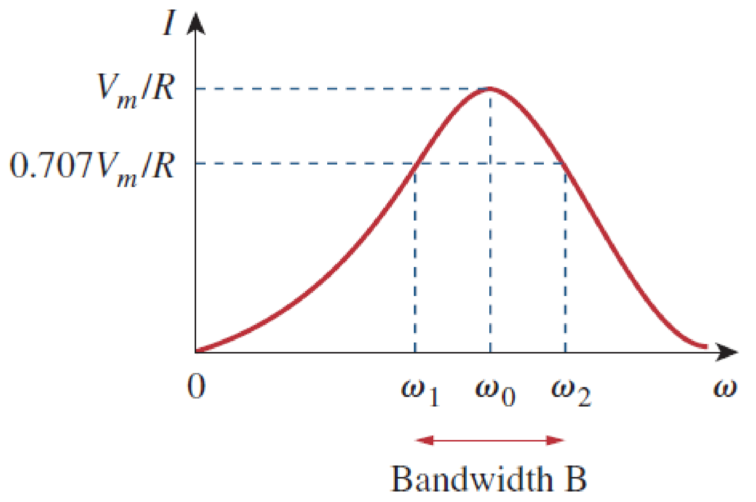
Series Resonance (II)

- Note that at resonance:
  1. The impedance is purely resistive, thus, $\mathbf{Z} = R$. In other words, the LC series combination acts like a short circuit, and the entire voltage is across R.
  2. The voltage $\mathbf{V}_s$ and the current $\mathbf{I}$ are in phase, so that the power factor is unity.
  3. The magnitude of the transfer function $\mathbf{H}(\omega) = \mathbf{Z}(\omega)$ is minimum.
  4. The inductor voltage and capacitor voltage can be much more than the source voltage.



- The frequency response of the circuit's current magnitude is

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$



- The highest power dissipated occurs at resonance, when $I = V_m/R$

$$P(\omega) = \frac{1}{2} I^2 R \quad \Rightarrow \quad P(\omega_0) = \frac{1}{2}\frac{V_m^2}{R}$$

Series Resonance (III)

- The frequency response of the circuit's current magnitude is

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

- At certain frequencies $\omega = \omega_1, \omega_2$, the dissipated power is half the maximum value; that is

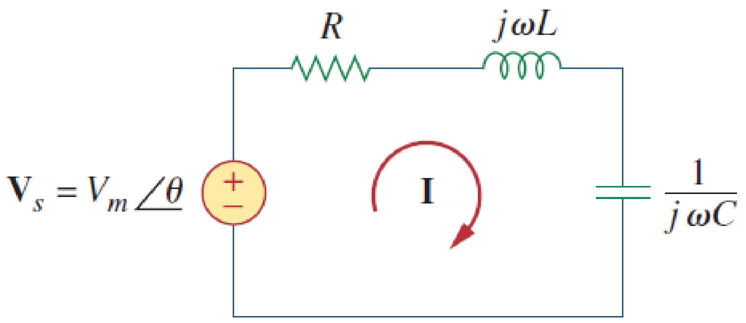
$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$

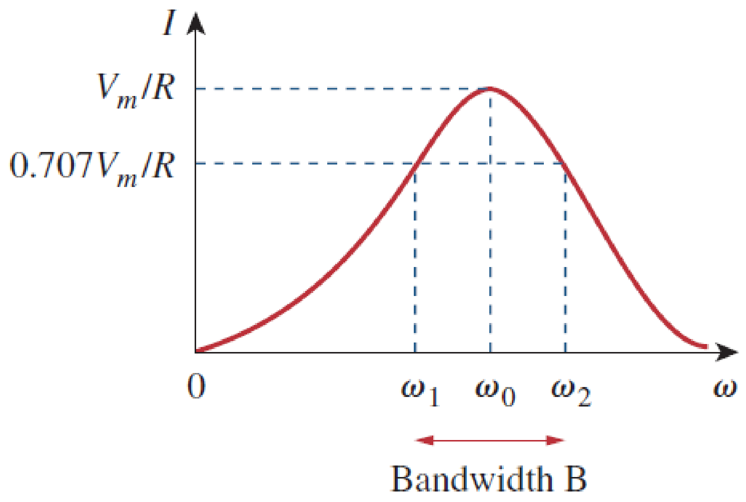
- The corresponding half-power frequencies (半功率频率) are

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \qquad \omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

- The width of the response curve depends on the bandwidth B (带宽), which is defined as the difference between the two half-power frequencies

$$B = \omega_2 - \omega_1$$





$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

Quality Factor (I)

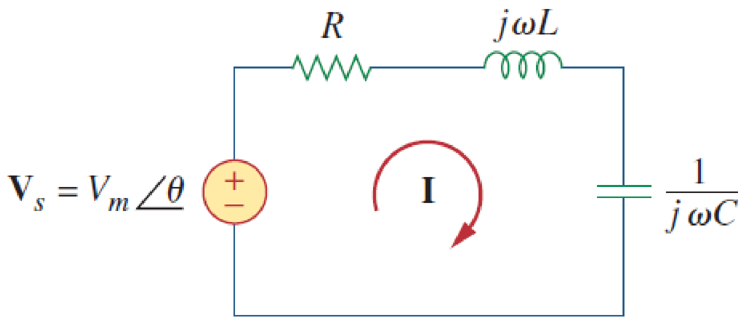
- We use quality factor Q (品质因数) to quantitively measure the "shapeness" of the resonance in a resonant circuit
- Quality factor is defined as the ratio between the peak energy stored in the circuit and the energy dissipated by the resistor in one period of time at resonance

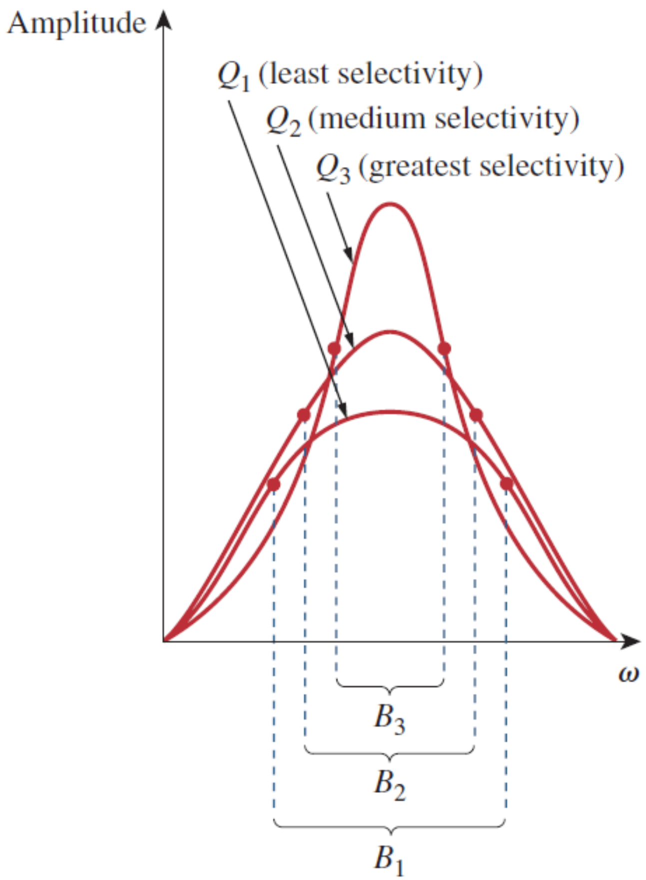
$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit} \atop \text{in one period at resonance}}$$

Series *RLC* Circuits

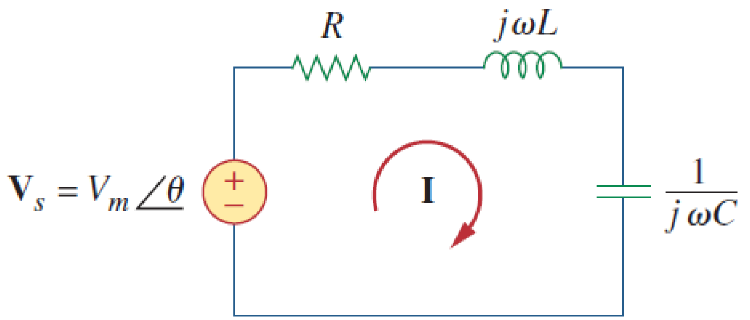
$$\boxed{Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \quad \text{or} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}}$$

- A higher Q value gives rise to shaper response curve, and so, better selectivity





Quality Factor (II)



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  Series *RLC* Circuits

$$\boxed{Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \quad \text{or} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}}$$

- The bandwidth B can be rewritten as

$$\boxed{B = \frac{R}{L} = \frac{\omega_0}{Q}}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

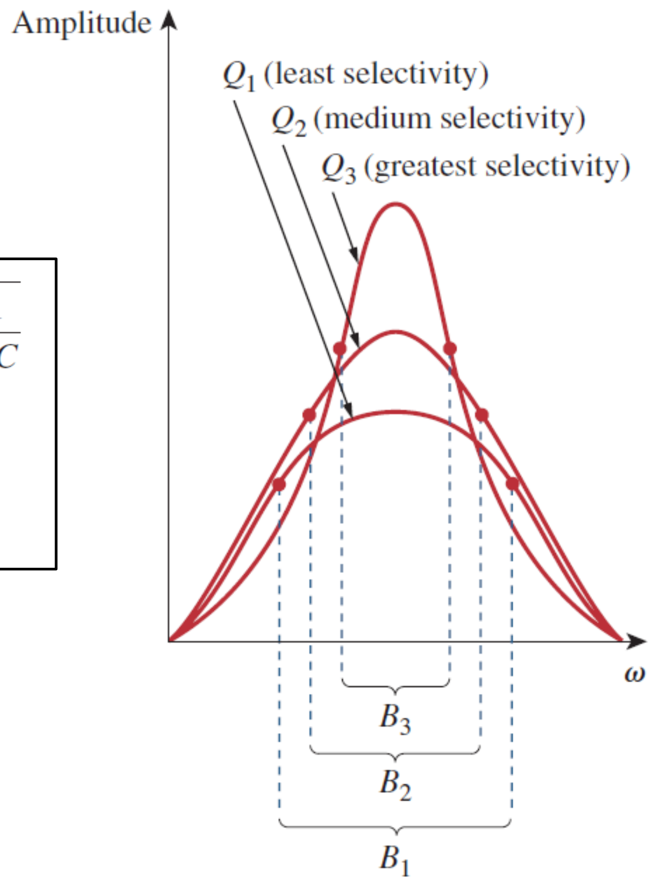
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_2 - \omega_1$$

  - A large Q leads to a narrow bandwidth

- For high-Q circuit ($Q > 10$), the half-power frequencies are symmetrical around the resonant frequency and can be estimated as

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2}$$



Example: Series Resonance (I)

In the circuit of Fig. 14.24, $R = 2\ \Omega$, $L = 1$ mH, and $C = 0.4\ \mu$F. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0, ω_1, and ω_2.

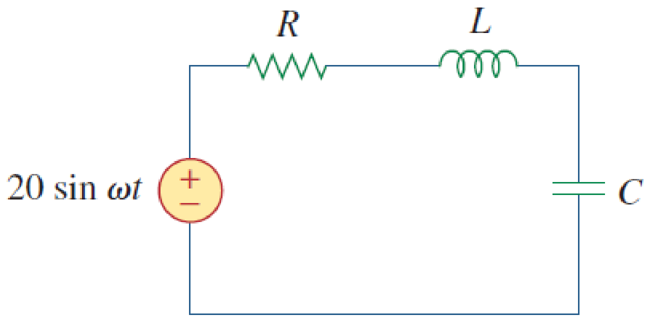


Figure 14.24
For Example 14.7.

Solution:

(a) The resonant frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$$

METHOD 1 The lower half-power frequency is

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$
$$= -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2}$$
$$= -1 + \sqrt{1 + 2500} \text{ krad/s} = 49 \text{ krad/s}$$

Similarly, the upper half-power frequency is

$$\omega_2 = 1 + \sqrt{1 + 2500} \text{ krad/s} = 51 \text{ krad/s}$$

METHOD 2 Alternatively, we could find

$$Q = \frac{\omega_0 L}{R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25$$

From Q, we find

$$B = \frac{\omega_0}{Q} = \frac{50 \times 10^3}{25} = 2 \text{ krad/s}$$

Since $Q > 10$, this is a high-Q circuit and we can obtain the half-power frequencies as

$$\omega_1 = \omega_0 - \frac{B}{2} = 50 - 1 = 49 \text{ krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 50 + 1 = 51 \text{ krad/s}$$

Example: Series Resonance (II)

In the circuit of Fig. 14.24, $R = 2\ \Omega$, $L = 1$ mH, and $C = 0.4\ \mu$F. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0, ω_1, and ω_2.

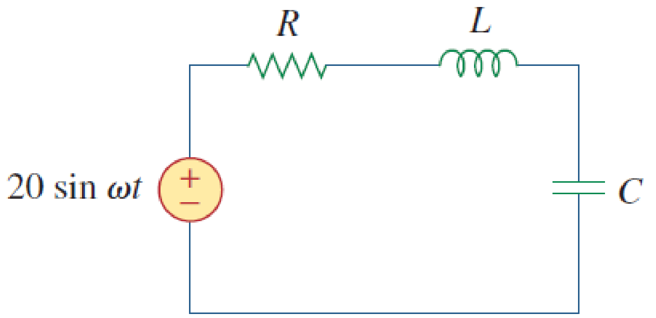


Figure 14.24
For Example 14.7.

(b) The bandwidth is

$$B = \omega_2 - \omega_1 = 2 \text{ krad/s}$$

or

$$B = \frac{R}{L} = \frac{2}{10^{-3}} = 2 \text{ krad/s}$$

The quality factor is

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

(c) At $\omega = \omega_0$,

$$I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$$

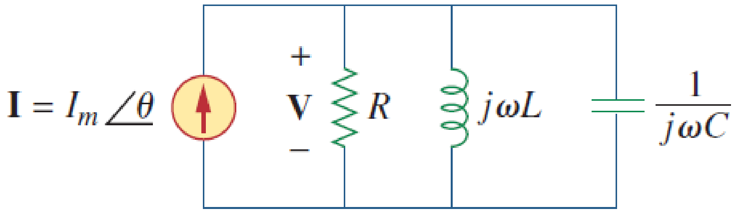
At $\omega = \omega_1, \omega_2$,

$$I = \frac{V_m}{\sqrt{2}R} = \frac{10}{\sqrt{2}} = 7.071 \text{ A}$$

Parallel Resonance (I)

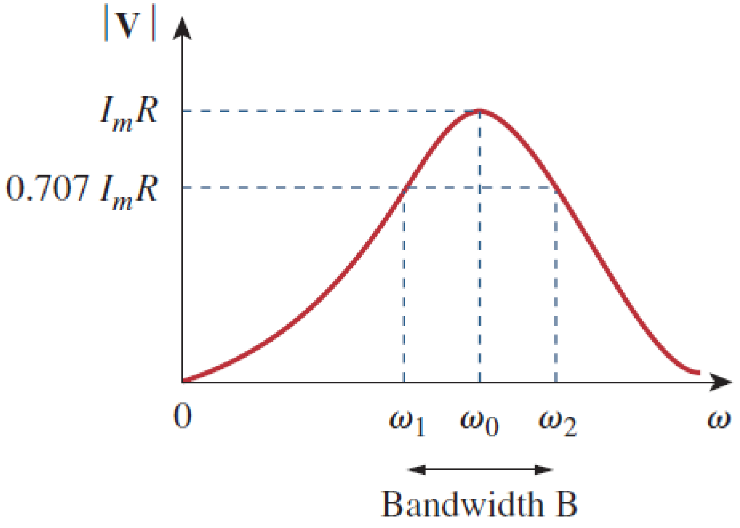
- The parallel *RLC* circuit is the dual of the series *RLC* circuit. The admittance is

$$\mathbf{Y} = H(\omega) = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$



- Resonance occurs when the imaginary part **Y** is zero

$$\omega C - \frac{1}{\omega L} = 0$$



- The resonant frequency is

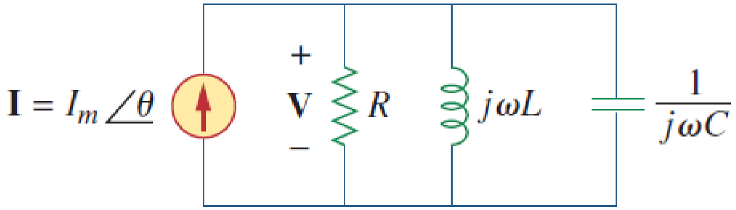
$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

- The half-power frequencies are

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

Parallel Resonance (II)



- The resonant frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

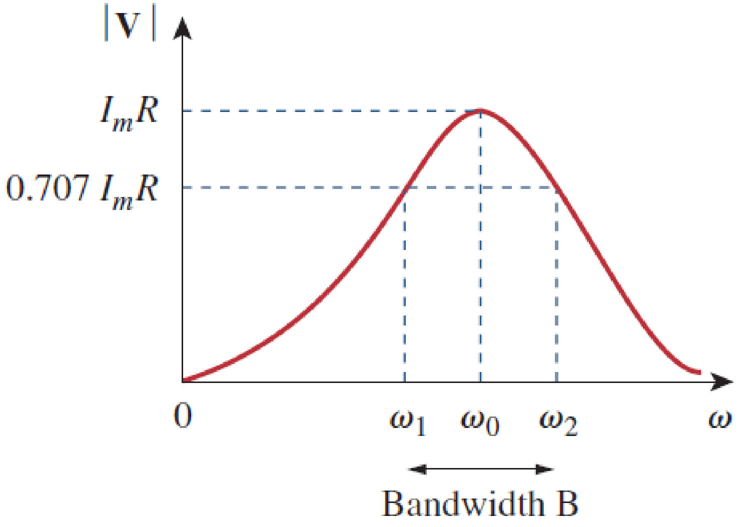
- The bandwidth

$$B = \omega_2 - \omega_1 = \frac{1}{RC}$$

- The quality factor

Parallel *RLC* Circuits

$$\boxed{Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L}}$$



- The half-power frequencies

$$\omega_1 = \omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_0}{2Q}, \qquad \omega_2 = \omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_0}{2Q}$$

$$\boxed{\begin{aligned} \omega_1 &= -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \end{aligned}}$$

- Again, for high-Q circuit ($Q \geq 10$)

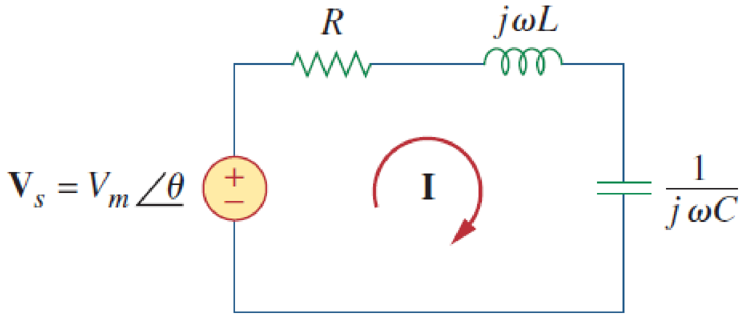
$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

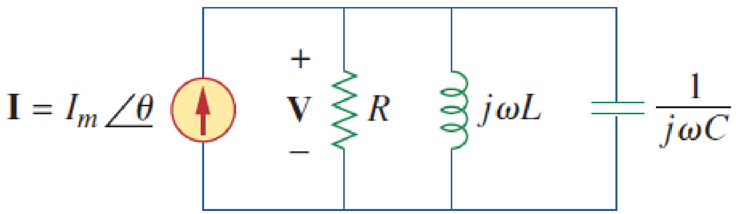
Summary: Resonant RLC Circuits

TABLE 14.4

Summary of the characteristics of resonant *RLC* circuits.

Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$





Example: Parallel Resonance

In the parallel *RLC* circuit of Fig. 14.27, let $R = 8\ \text{k}\Omega$, $L = 0.2$ mH, and $C = 8\ \mu$F. (a) Calculate ω_0, Q, and B. (b) Find ω_1 and ω_2. (c) Determine the power dissipated at ω_0, ω_1, and ω_2.

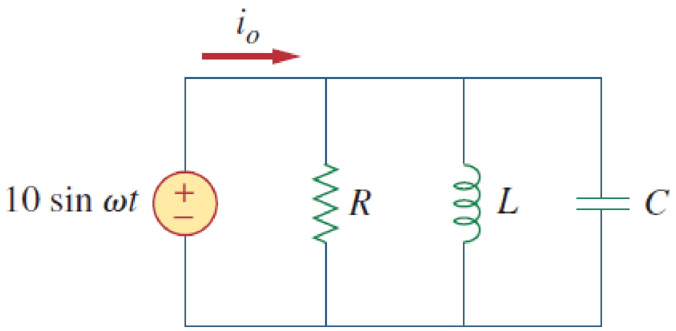


Figure 14.27
For Example 14.8.

Solution:

(a)

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2 \times 10^{-3} \times 8 \times 10^{-6}}} = \frac{10^5}{4} = 25\ \text{krad/s}$$

$$Q = \frac{R}{\omega_0 L} = \frac{8 \times 10^3}{25 \times 10^3 \times 0.2 \times 10^{-3}} = 1{,}600$$

$$B = \frac{\omega_0}{Q} = 15.625\ \text{rad/s}$$

(b) Due to the high value of Q, we can regard this as a high-Q circuit, Hence,

$$\omega_1 = \omega_0 - \frac{B}{2} = 25{,}000 - 7.812 = 24{,}992\ \text{rad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 25{,}000 + 7.812 = 25{,}008\ \text{rad/s}$$

(c) At $\omega = \omega_0$, $\mathbf{Y} = 1/R$ or $\mathbf{Z} = R = 8\ \text{k}\Omega$. Then

$$\mathbf{I}_o = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{10\angle -90^\circ}{8{,}000} = 1.25\angle -90^\circ\ \text{mA}$$

$$P = \frac{V_m^2}{2R} = \frac{100}{2 \times 8 \times 10^3} = 6.25\ \text{mW}$$

At $\omega = \omega_1, \omega_2$,

$$P = \frac{V_m^2}{4R} = 3.125\ \text{mW}$$

Outline

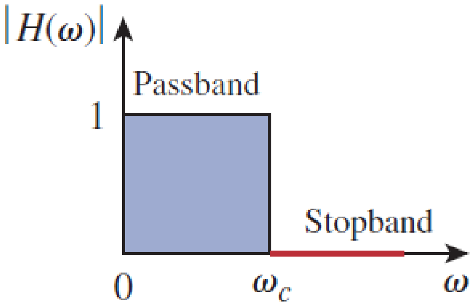
- Introduction
- Transfer Function
- Series Resonance
- Parallel Resonance
- **Passive Filters**
- **Active Filters**

Passive Filter

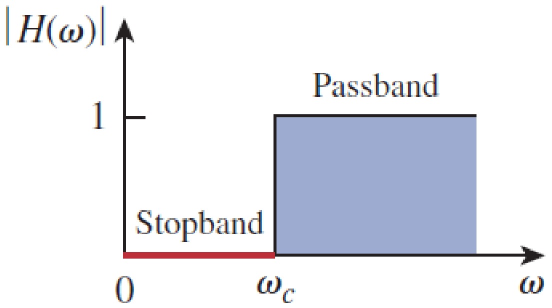
- A filter (滤波器) is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others

There are four types of filters

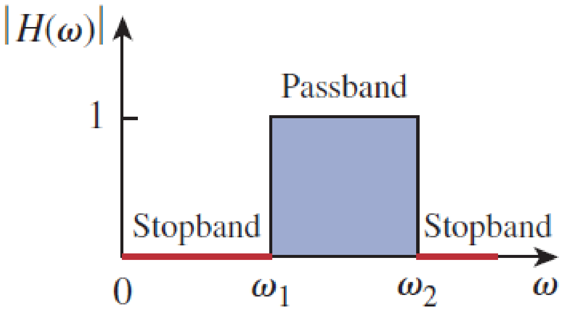
- Lowpass filter passes only low frequencies and blocks high frequencies
- Highpass filter does the opposite of lowpass
- Bandpass filter only allows a range of frequencies to pass through
- Bandstop filter does the opposite of bandpass



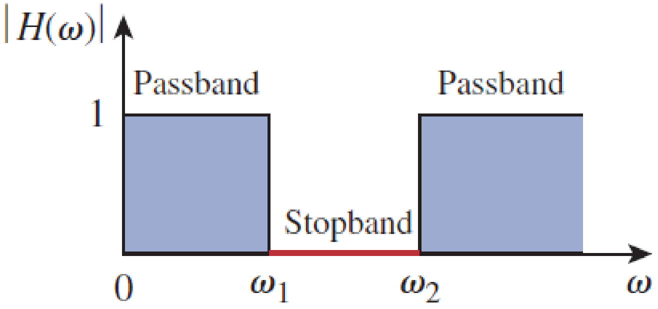
低通滤波器



高通滤波器



带通滤波器

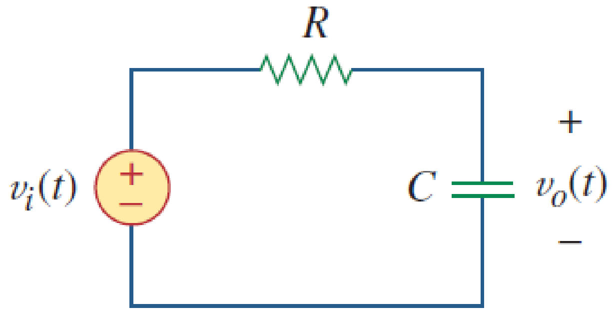


带阻滤波器

Passive Lowpass Filter

- A typical lowpass filter is formed when the output of a *RC* circuit is taken from the capacitor

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$$

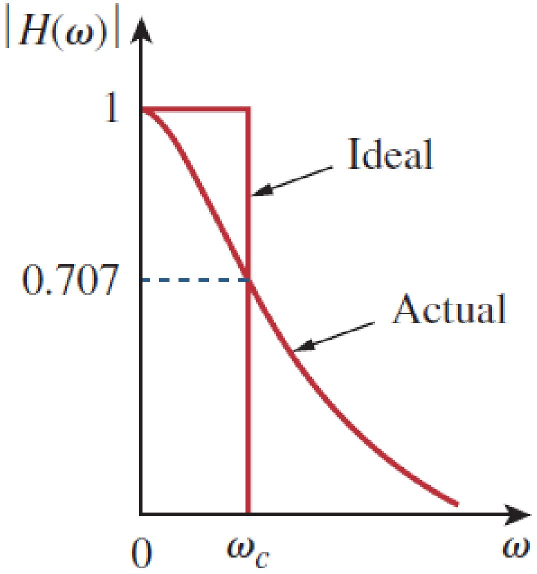


- The half power frequency (半功率频率) is

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{RC}$$

  - ➢ This is also referred to as the cutoff frequency (截止频率)

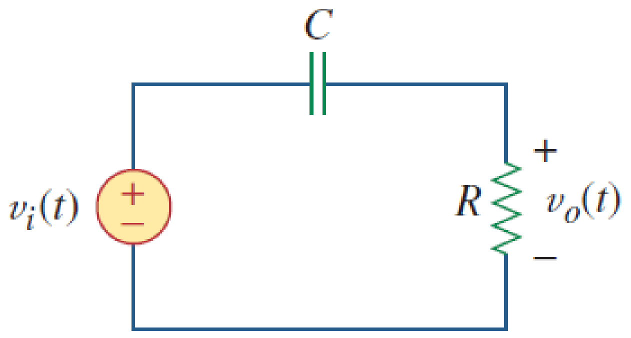
- The filter is designed to pass from DC up to ω_c



Passive Highpass Filter

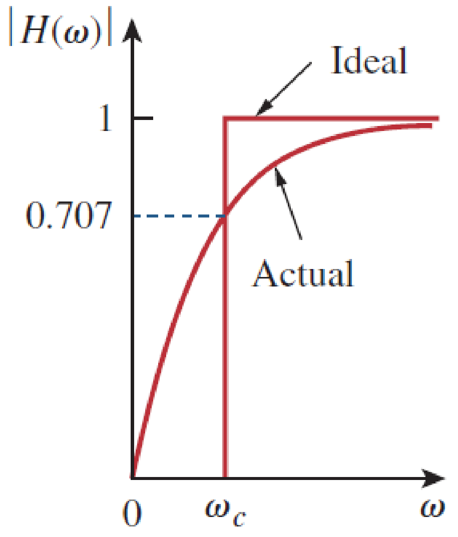
- A highpass filter is also made of a *RC* circuit, with the output taken off the resistor

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + 1/j\omega C} = \frac{j\omega RC}{1 + j\omega RC}$$



- The cutoff frequency is the same as the lowpass filter

$$\omega_c = \frac{1}{RC}$$

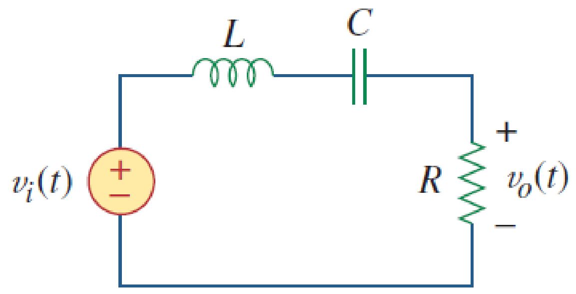


- The difference being that the frequencies passed go from ω_c to infinity

Passive Bandpass Filter

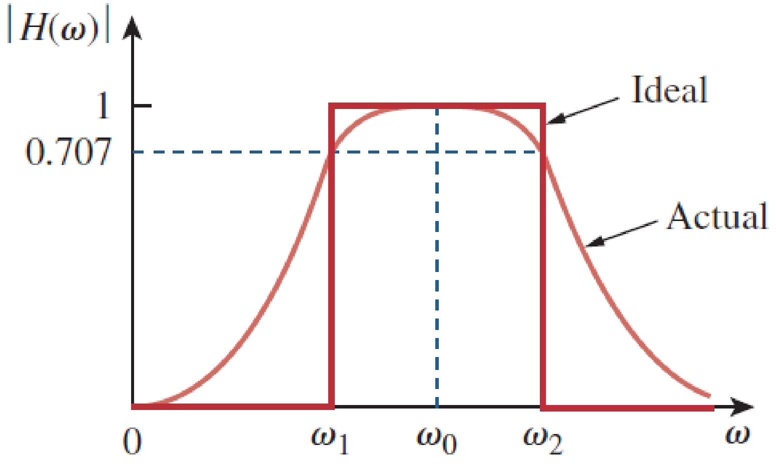
- The *RLC* series resonant circuit provides a bandpass filter when the output is taken off the resistor

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$



- The center frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

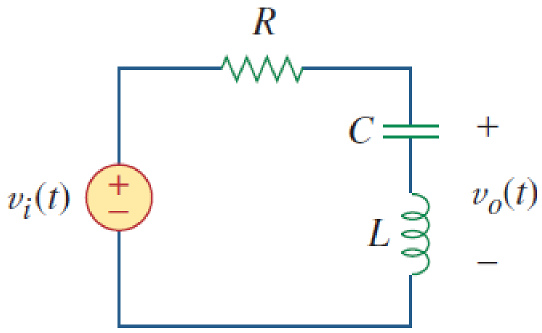


- The filter will pass frequencies from ω_1 to ω_2
  - ➢ ω_1: lower cut-off frequency (下限截止频率)
  - ➢ ω_2: upper cut-off frequency (上限截止频率)

Passive Bandreject Filter

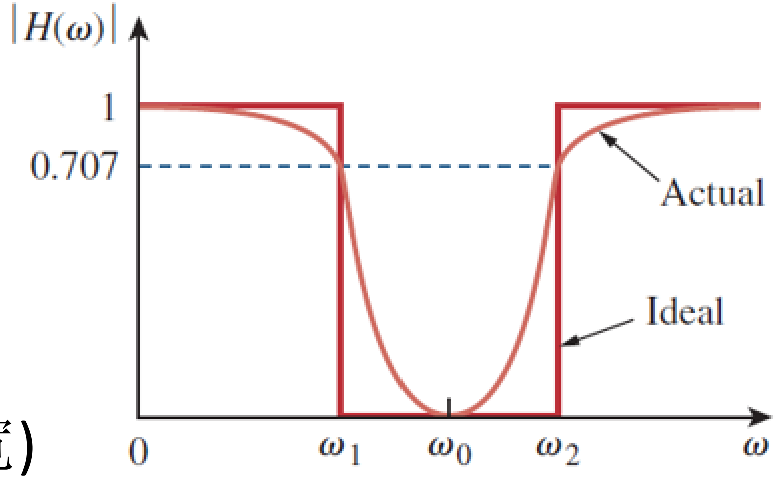
- A bandreject filter can be created from a *RLC* circuit by taking the output from the *LC* series combination

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$



- The center frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



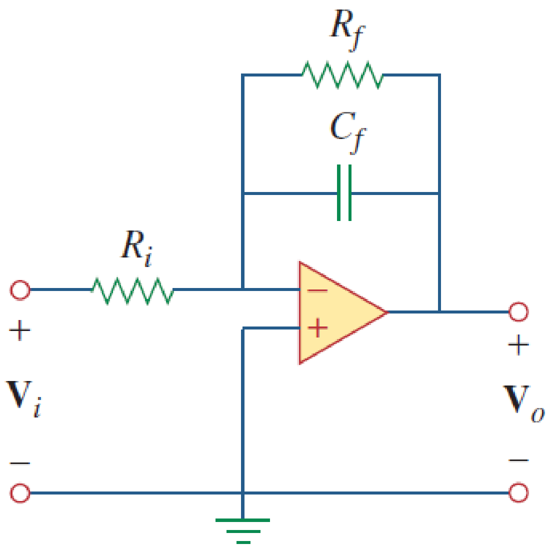
- The filter will pass frequencies from ω_1 to ω_2
  - ➢ ω_0 is called frequency of rejection (抑制频率)
  - ➢ $B = \omega_2 - \omega_1$ is called bandwidth of rejection (抑制带宽)

Active Filters

- Limitations of passive filters:
  - Gain ≤ 1
  - May need to use bulky, costly inductors; Hard to integrate in integrated circuits (ICs)
  - Poor performance at low frequency range (300 Hz < f < 3 kHz)
- Passive filters are mostly used for high frequency circuits, such as in telecommunications
- Active filter can address those limitations with advantages of small form factor, suitable for integration on IC, flexible gain and isolation between stages allowing for independent and optimal design of each stage
  - Limited by the bandwidth of op amp, it commonly used below frequency 100 kHz

Active Lowpass and High Pass Filter

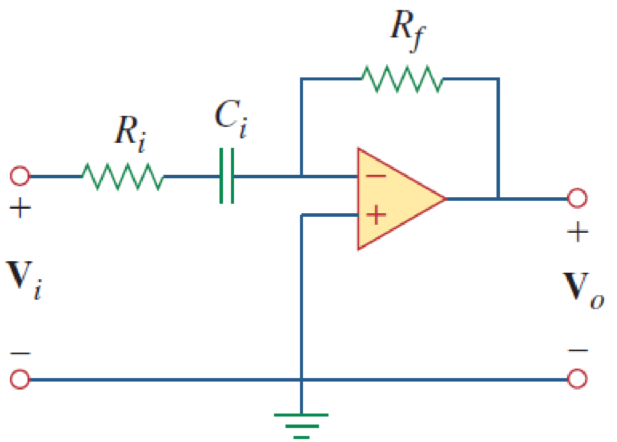
Active first-order lowpass filter



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{R_f}{R_i}\frac{1}{1 + j\omega C_f R_f}$$

$$\omega_c = \frac{1}{R_f C_f}$$

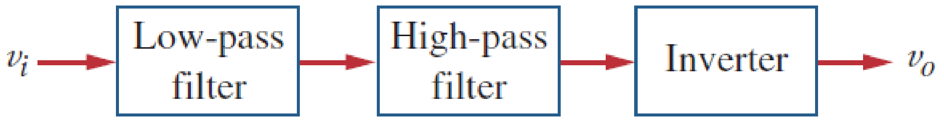
Active first-order highpass filter

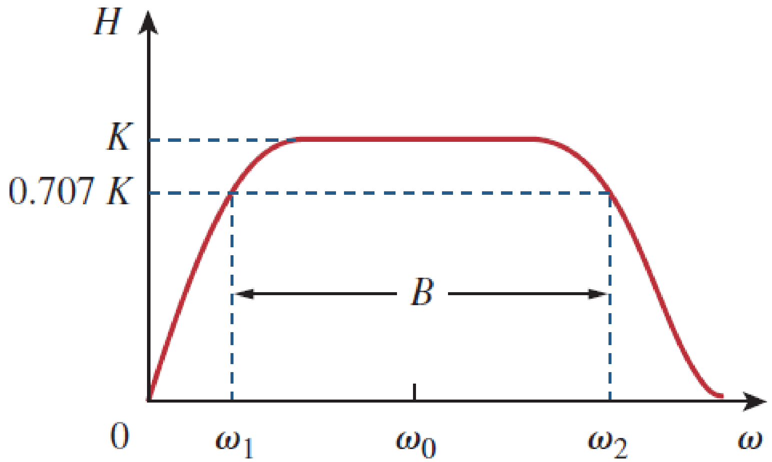


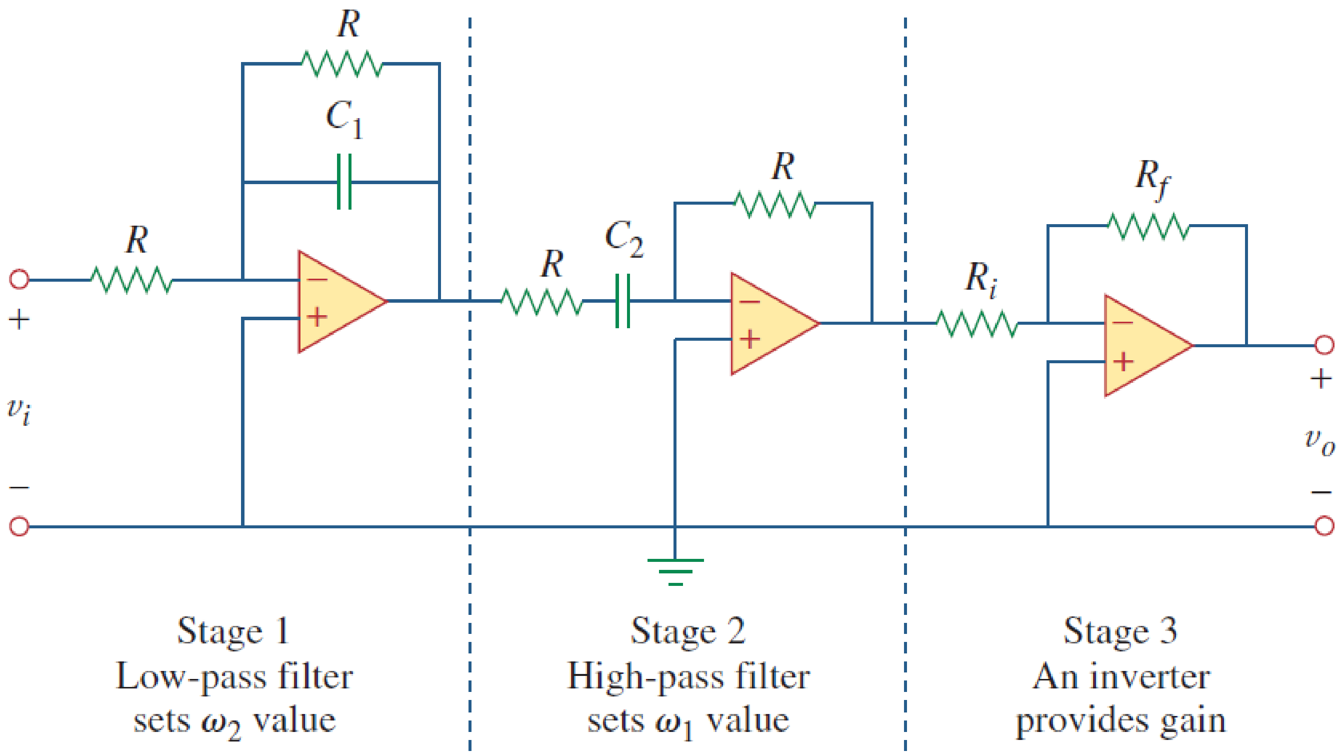
$$\mathbf{H}(\omega) = -\frac{R_f}{R_i + 1/j\omega C_i} = -\frac{j\omega C_i R_f}{1 + j\omega C_i R_i}$$

$$\omega_c = \frac{1}{R_i C_i}$$

Active Bandpass Filter







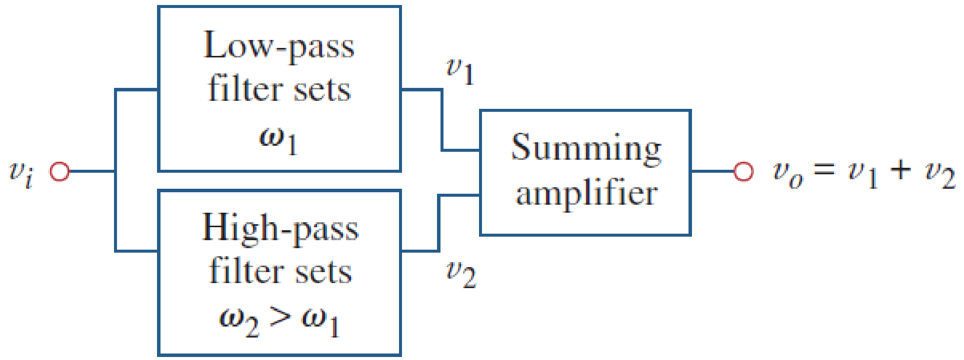
$$\omega_2 = \frac{1}{RC_1} \qquad \omega_1 = \frac{1}{RC_2} \qquad \omega_0 = \sqrt{\omega_1 \omega_2}$$

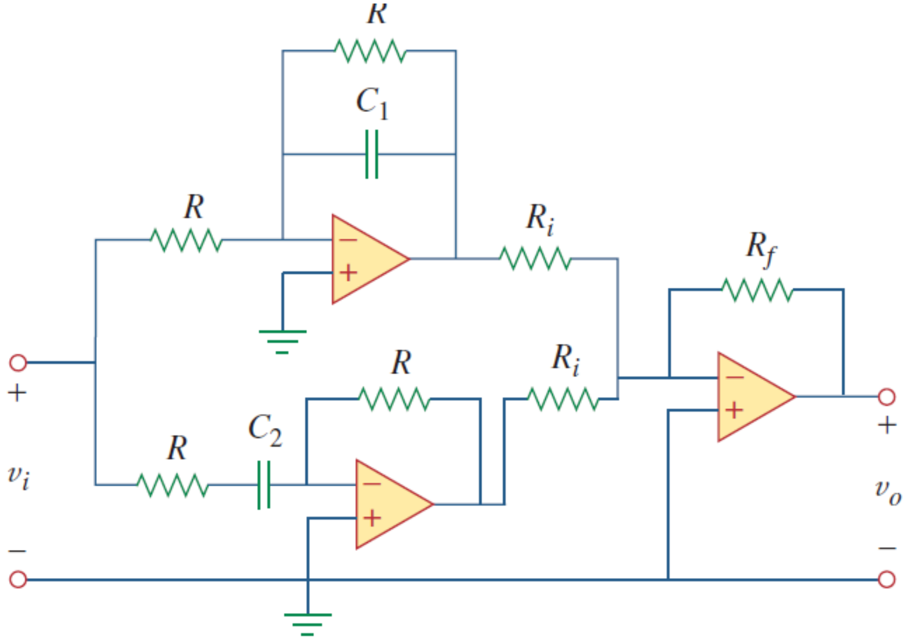
$$B = \omega_2 - \omega_1$$

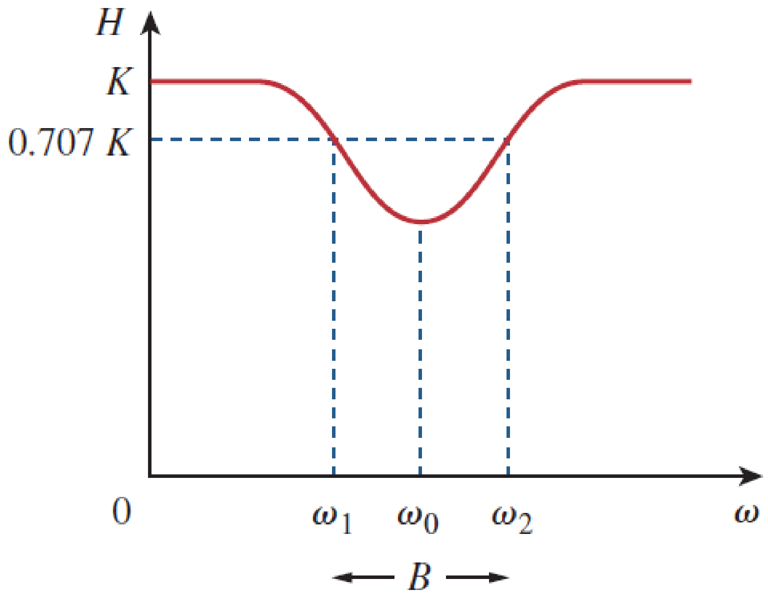
$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R}\right)\left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R}\right)\left(-\frac{R_f}{R_i}\right)$$

$$= -\frac{R_f}{R_i}\,\frac{1}{1 + j\omega C_1 R}\,\frac{j\omega C_2 R}{1 + j\omega C_2 R}$$

Active Bandreject (Notch) Filter







$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{R_f}{R_i}\left(-\frac{1}{1 + j\omega C_1 R} - \frac{j\omega C_2 R}{1 + j\omega C_2 R}\right)$$

$$\omega_1 = \frac{1}{RC_1} \qquad \omega_2 = \frac{1}{RC_2}$$

Summary

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